KAIZEN MATHS

10-MINUTE PRACTICE

Algebraic substitution

Quadratic expressions

Substitute into squared terms, then follow the order of operations.

10 MINUTES / 6 QUESTIONS / AS-02

Name _____ Date _____

Show your working. Use brackets for negative values.

1 Given $x = 5$

Evaluate $x^2 + 2x$

2 Given $x = -3$

Evaluate $x^2 + 3x + 5$

3 Given $x = -4$

Evaluate $x^2 + 3x - 1$

4 Given $x = -1$

Evaluate $x^2 + 5x - 8$

5 Given $x = 3$

Evaluate $x^2 + 4x - 7$

6 Given $x = -4$

Evaluate $x^2 + 2x + 6$

Answer sheet

Quadratic expressions | 6 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 35

02 5

03 3

04 -12

05 14

06 14

Worked solutions

Quadratic expressions

01 Given $x = 5$

$x^2 + 2x$

Substitute $x = 5$.

$(5)^2 + 2\times 5$

$25 + 10$

35

02 Given $x = -3$

$x^2 + 3x + 5$

Substitute $x = -3$.

$(-3)^2 + 3\times(-3) + 5$

$9 - 9 + 5$

5

03 Given $x = -4$

$x^2 + 3x - 1$

Substitute $x = -4$.

$(-4)^2 + 3\times(-4) - 1$

$16 - 12 - 1$

3

04 Given $x = -1$

$x^2 + 5x - 8$

Substitute $x = -1$.

$(-1)^2 + 5\times(-1) - 8$

$1 - 5 - 8$

-12

Worked solutions

Quadratic expressions

05 Given $x = 3$

$x^2 + 4x - 7$

Substitute $x = 3$.

$(3)^2 + 4\times 3 - 7$

$9 + 12 - 7$

14

06 Given $x = -4$

$x^2 + 2x + 6$

Substitute $x = -4$.

$(-4)^2 + 2\times (-4) + 6$

$16 - 8 + 6$

14